Exercises: Topological Basics and Continuity

Mathematical Foundations for Machine Learning

Difficulty: $\star$ Easy $\star\star$ Medium $\star\star\star$ Hard

1 Open and Closed Sets

Exercise 1.1 ($\star$). *Let $\Omega = (0,1) \subset \mathbb{R}$. Using the definition of an open set, prove that Ω is open. That is, show that for every $x \in (0,1)$ there exists $\varepsilon > 0$ such that $B(x,\varepsilon) \subset (0,1)$.*

Exercise 1.2 ($\star$). *Determine whether the following subsets of $\mathbb{R}^2$ (with the Euclidean norm) are open, closed, both, or neither. Justify each answer.*

(a) $A = \{(x,y) \in \mathbb{R}^2 : x^2 + y^2 < 1\}$

(b) $B = \{(x,y) \in \mathbb{R}^2 : x^2 + y^2 \leq 1\}$

(c) $C = \{(x,y) \in \mathbb{R}^2 : 0 < x^2 + y^2 \leq 1\}$

(d) $D = \mathbb{R}^2$

Exercise 1.3 ($\star\star$). *Prove that the set $\Omega = \{(x,y) \in \mathbb{R}^2 : x > 0 \text{ and } y > 0\}$ (the open first quadrant) is an open set using the definition with open balls.*

Exercise 1.4 ($\star\star$). *Let $\Omega = [0,1] \subset \mathbb{R}$. Using the sequential characterization of closed sets (every convergent sequence with elements in Ω has its limit in Ω), prove that Ω is closed.*

Exercise 1.5 ($\star\star\star$). *Prove that a finite intersection of open sets is open, and that an arbitrary union of open sets is open. Give an example showing that an infinite intersection of open sets need not be open.*

2 Closure, Interior, and Boundary

Exercise 2.1 ($\star$). *For the set $\Omega = (0,1] \subset \mathbb{R}$, determine:*

(a) The closure $\overline{\Omega}$

(b) The interior Ω^o

(c) The boundary $\partial\Omega$

Verify that $\partial\Omega = \overline{\Omega} \setminus \Omega^o$.

Exercise 2.2 ($\star$). *For $\Omega = \mathbb{Q} \cap [0,1]$ (the rationals in $[0,1]$), find $\overline{\Omega}$, Ω^o, and $\partial\Omega$.*

Exercise 2.3 ($\star\star$). *Let $\Omega = \{(x,y) \in \mathbb{R}^2 : x^2 + y^2 < 1\} \cup \{(1,0)\}$. Find the closure, interior, and boundary of Ω.*

Exercise 2.4 ($\star\star$). *Prove that for any set $\Omega \subset \mathbb{R}^n$:*

(a) Ω^o *is open.*

(b) $\overline{\Omega}$ *is closed.*

(c) Ω *is open if and only if* $\Omega = \Omega^o$.

Exercise 2.5 ($\star\star\star$). *Let $A, B \subset \mathbb{R}^n$. Prove or give a counterexample:*

(a) $\overline{A \cup B} = \overline{A} \cup \overline{B}$

(b) $\overline{A \cap B} = \overline{A} \cap \overline{B}$

(c) $(A \cap B)^o = A^o \cap B^o$

(d) $(A \cup B)^o = A^o \cup B^o$

3 Boundedness, Compactness, and Bolzano–Weierstrass

Exercise 3.1 ($\star$). *Determine which of the following subsets of $\mathbb{R}^n$ are bounded. Justify your answers.*

(a) $A = \{x \in \mathbb{R}^2 : \|x\|_2 \leq 5\}$

(b) $B = \{x \in \mathbb{R}^2 : x_1^2 - x_2^2 < 1\}$

(c) $C = \{x \in \mathbb{R}^n : \|x\|_1 < 10\}$

Exercise 3.2 ($\star$). *Determine which of the following sets are compact in $\mathbb{R}^n$. Justify using Theorem 52 (compact $\Leftrightarrow$ closed and bounded in $\mathbb{R}^n$).*

(a) $[0,1]^n$

(b) $(0,1)^n$

(c) $\{x \in \mathbb{R}^n : \|x\|_2 = 1\}$ *(the unit sphere)*

(d) $\{1/n : n \in \mathbb{N}\}$

(e) $\{0\} \cup \{1/n : n \in \mathbb{N}\}$

Exercise 3.3 ($\star\star$). *Let $(x_k)_{k\in\mathbb{N}}$ be a sequence in $\mathbb{R}^2$ defined by $x_k = \left(\cos(k), \frac{(-1)^k}{k}\right)$. Show that this sequence is bounded and use the Bolzano–Weierstrass theorem to conclude that it has a convergent subsequence.*

Exercise 3.4 ($\star\star$). *Let $K \subset \mathbb{R}^n$ be compact and let (x_k) be a sequence in K such that every convergent subsequence converges to the same limit $x^* \in K$. Prove that the full sequence (x_k) converges to x^*.*

Exercise 3.5 ($\star\star\star$). *Let $K_1 \supset K_2 \supset K_3 \supset \cdots$ be a nested sequence of non-empty compact sets in $\mathbb{R}^n$. Prove that $\bigcap_{i=1}^{\infty} K_i \neq \emptyset$. Show by example that the conclusion can fail if the sets are only closed (but not bounded).*

4 Continuity

Exercise 4.1 ($\star$). *Using the ε-δ definition, prove that $f : \mathbb{R} \to \mathbb{R}$ defined by $f(x) = 3x + 2$ is continuous.*

Exercise 4.2 ($\star$). *Let $f : \mathbb{R}^n \to \mathbb{R}$ be defined by $f(x) = \|x\|_2$. Prove that f is continuous. Hint: use the reverse triangle inequality.*

Exercise 4.3 ($\star\star$). *Prove using the preimage characterization (Proposition 3) that the set $\{x \in \mathbb{R}^n : \|x\|_2 < 1\}$ is open and the set $\{x \in \mathbb{R}^n : \|x\|_2 \leq 1\}$ is closed.*

Exercise 4.4 ($\star\star$). *Let $f : \mathbb{R}^n \to \mathbb{R}^m$ be continuous and let $U \subset \mathbb{R}^n$ be open. Is $f(U)$ necessarily open? Prove or give a counterexample.*

Exercise 4.5 ($\star\star\star$). *Let $f : K \to \mathbb{R}$ be continuous on a compact set $K \subset \mathbb{R}^n$. Prove that f is uniformly continuous on K, i.e., for every $\varepsilon > 0$ there exists $\delta > 0$ such that $\|f(x) - f(y)\| < \varepsilon$ for all $x, y \in K$ with $\|x - y\| < \delta$.*

5 Extreme Value Theorem and Distance to Sets

Exercise 5.1 ($\star$). *Let $f(x) = x^2 - 2x + 3$ on $K = [0, 3]$. Find the minimum and maximum of f on K and state which theorem guarantees their existence.*

Exercise 5.2 ($\star\star$). *Let $K = \{x \in \mathbb{R}^2 : \|x\|_2 \leq 1\}$ and $f(x) = x_1^2 + 2x_2^2 - x_1$. Find the minimum and maximum of f on K and verify that Corollary 7 applies.*

Exercise 5.3 ($\star\star$). *Let $K = \{x \in \mathbb{R}^2 : x_1 \geq 0,\ x_2 \geq 0,\ x_1 + x_2 \leq 1\}$ (a triangle). Show that K is compact, and for $p = (2, 2)$, find $d(p, K) = \inf_{u \in K} \|p - u\|_2$ and the point in K that attains it.*

Exercise 5.4 ($\star\star\star$). *Let $M \subset \mathbb{R}^n$ be a non-empty closed set and $x \in \mathbb{R}^n$. Prove that there exists $m^* \in M$ such that $\|x - m^*\| = d(x, M) := \inf_{u \in M} \|x - u\|$. Hint: reduce to optimization over a compact set as described in the notes.*

6 Path-Connectedness

Exercise 6.1 ($\star$). *Show that every convex set $C \subset \mathbb{R}^n$ is path-connected. Recall that C is convex if for all $x, y \in C$ and $t \in [0, 1]$, $(1 - t)x + ty \in C$.*

Exercise 6.2 ($\star\star$). *Show that the set $\Omega = \mathbb{R}^2 \setminus \{(0, 0)\}$ is path-connected.*

Exercise 6.3 ($\star\star$). *Prove that if $f : V_1 \to V_2$ is continuous and $M \subset V_1$ is path-connected, then $f(M)$ is path-connected.*

Exercise 6.4 ($\star\star\star$). *Consider the set of invertible $n \times n$ real matrices $\mathrm{GL}_n(\mathbb{R}) = \{A \in \mathbb{R}^{n \times n} : \det(A) \neq 0\}$.*

(a) Prove that $\mathrm{GL}_n(\mathbb{R})$ is an open set in $\mathbb{R}^{n \times n}$.

(b) Prove that $\mathrm{GL}_n(\mathbb{R})$ has exactly two path-connected components: $\{A : \det(A) > 0\}$ and $\{A : \det(A) < 0\}$.

7 Mixed Challenging Exercises

Exercise 7.1 ($\star\star\star$). *Let $f : \mathbb{R}^n \to \mathbb{R}$ be continuous. Prove that for every $c \in \mathbb{R}$, the sublevel set $S_c = \{x \in \mathbb{R}^n : f(x) \leq c\}$ is closed. Explain why this is relevant in optimization (e.g., for defining feasible sets).*

Exercise 7.2 ($\star\star\star$). *Let $K \subset \mathbb{R}^n$ be compact and non-empty. Prove that K has a finite diameter, i.e., $\operatorname{diam}(K) = \sup_{x,y \in K} \|x - y\| < \infty$, and that the supremum is attained.*

Exercise 7.3 ($\star\star\star$). *Let (f_k) be a sequence of continuous functions $f_k : K \to \mathbb{R}$ on a compact set $K \subset \mathbb{R}^n$ that converges uniformly to $f : K \to \mathbb{R}$. Prove that f is continuous. Provide a counterexample showing that pointwise convergence is not sufficient.*

Exercise 7.4 ($\star\star\star$). *Let $A \in \mathbb{R}^{m \times n}$ and define $f : \mathbb{R}^n \to \mathbb{R}^m$ by $f(x) = Ax$. Prove that:*

(a) f is continuous.

(b) If $C \subset \mathbb{R}^n$ is compact, then $f(C)$ is compact.

(c) If $M \subset \mathbb{R}^n$ is path-connected, then $f(M)$ is path-connected.

(d) The null space $\ker(A) = \{x : Ax = 0\}$ is closed.

Exercise 7.5 ($\star\star\star$). ***(Application to ML)*** *Consider a neural network layer $f(x) = \sigma(Wx+b)$ where σ is a continuous activation function (e.g., ReLU, sigmoid), $W \in \mathbb{R}^{m \times n}$, and $b \in \mathbb{R}^m$.*

(a) Prove that f is continuous.

(b) If the input is constrained to a compact set K (e.g., normalized pixel values in $[0, 1]^n$), what can you conclude about the range $f(K)$?

(c) Explain why path-connectedness of the loss landscape over weight space matters for gradient-based optimization.